

| TestID              | Description                         | Input                                                                                                                              | Expected Output                                                                                                | Actual Output                                                                                                  | Pass/Fail |
|---------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------|
| RegistrationTest_01 | Registration with valid details     | Name: John Doe<br>Email: john@example.com<br>Role: Student<br>Password: StrongP@ss<br>Token: STUDENTCB6TL                          | User should be registered successfully, no errors                                                              | User registered succesfully, no errors                                                                         | Pass      |
| RegistrationTest_02 | Registration with invalid name      | Name: 1234<br>Email: john@example.com<br>Role: Teacher<br>Password: StrongP@ss<br>Token: INSTRUCTD7IDD                             | Error message: "Invalid name. Please use letters only."                                                        | Error message: "Invalid name. Please use letters only."                                                        | Pass      |
| RegistrationTest_03 | Registration with invalid email     | Name: John Doe<br>Email: johnexample.com<br>Role: Admin<br>Password: StrongP@ss<br>Token: INSTRUCTD7IDD                            | Error message: "Invalid email format."                                                                         | Error message: "Invalid email format."                                                                         | Pass      |
| RegistrationTest_04 | Registration with invalid password  | Name: John Doe<br>Email: john@example.com<br>Role: Student<br>Password: 123 (too short, missing complexity)<br>Token: STUDENTCB6TL | Cannot create user!<br>FirebaseError: Firebase: Password should be at least 6 characters (auth/weak-password). | Cannot create user!<br>FirebaseError: Firebase: Password should be at least 6 characters (auth/weak-password). | Pass      |
|                     |                                     |                                                                                                                                    |                                                                                                                |                                                                                                                |           |
| LoginTest_01        | Login with valid email and password | Email: <a href="mailto:john@example.com">john@example.com</a><br>Password: StrongP@ss                                              | User is successfully logged in and redirected to dashboard                                                     | User is successfully logged in and redirected to dashboard                                                     | Pass      |

|              |                                           |                                                                                         |                                                                                                          |                                                                                                          |      |
|--------------|-------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|------|
| LoginTest_02 | Login with invalid email format           | Email: johnexample.com<br>Password: StrongP@ss                                          | Error message – “Invalid email format.”                                                                  | Error message – “Invalid email format.”                                                                  | Pass |
| LoginTest_03 | Login with invalid password format        | Email: <a href="mailto:john@example.com">john@example.com</a><br>Password: 12345        | Error message – “Cannot sign in user! FirebaseError: Firebase: Error (auth/invalid-login-credentials).”  | Error message – “Cannot sign in user! FirebaseError: Firebase: Error (auth/invalid-login-credentials).”  | Pass |
| LoginTest_04 | Login with blank fields                   | Email: [empty]<br>Password: [empty]                                                     | Error message – “Cannot sign in user! FirebaseError: Firebase: Error (auth/invalid-login-credentials)..” | Error message – “Cannot sign in user! FirebaseError: Firebase: Error (auth/invalid-login-credentials)..” | Pass |
| LoginTest_05 | Login with valid email but wrong password | Email: <a href="mailto:john@example.com">john@example.com</a><br>Password: WrongPass123 | Error message – “Cannot sign in user! FirebaseError: Firebase: Error (auth/invalid-login-credentials).”  | Error message – “Cannot sign in user! FirebaseError: Firebase: Error (auth/invalid-login-credentials)..” | Pass |